

AYOUB ALWULAYI

Computer Engineering Student

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SUMMARY

Computer Engineering student At **Al Qassim University** with hands-on experience in backend development, software engineering, and embedded systems. Proficient in JavaScript/Node.js, REST API design, and low-level C/C++ programming. Experienced with Git/GitHub for version control and collaborative development. Seeking a **Summer internship** in Software Engineering or Backend Development.

TECHNICAL SKILLS

Languages: JavaScript, C++, C, Python, SQL, HTML, CSS, MATLAB, Verilog, irvine Assembly

Backend & Web: Node.js, REST API, Express.js, SQL Databases, JSON

Tools & DevOps: Git, GitHub, VS Code

Embedded & Hardware: ATmega328P, Arduino, I2C, SPI, PWM, ADC, Interrupts, Proteus Simulation

FPGA & Digital Design: Xilinx Vivado, Xilinx ISE, Verilog HDL, FSMs, Timing Analysis

Software Concepts: Software Engineering, System Engineering, OOP, Data Structures & Algorithms, Computer Architecture

PROJECTS

RESTful API & Backend System | [Personal Project](#) |

- Designed and implemented a backend server using Node.js and Express.js with RESTful API endpoints
- Managed codebase using Git/GitHub with structured branching, commits, and version control best practices
- Worked with SQL database integration for persistent data storage and retrieval

Intelligent Temperature Control System | [Embedded Systems Project](#) |

- Built a real-time monitoring system using LM35 sensor and 10-bit ADC with AVR C firmware
- Implemented hysteresis-based control logic for fan/heater management and LCD visualization
- Validated full system behavior through Proteus simulation before hardware deployment

Embedded LCD Interface System | [Low-Level Driver Project](#) |

- Developed low-level AVR C drivers for 16x2 LCD, achieving deterministic real-time performance
- Designed hardware schematics and simulated full system in Proteus
- Applied modular code structure and clean documentation practices for maintainability

LTI System Analysis | [MATLAB / Signal Processing Project](#) |

- Implemented convolution, Z-transform, and digital filter design in MATLAB
- Conducted stability and frequency-domain analysis for signal processing applications

EDUCATION

Bachelor of Computer Engineering | Graduation: *Next Semester*

- GPA: 3.97 / 5.0

LANGUAGES

Arabic (Native) **English** (Professional)